

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|------------------------------------|--|
| 1. Function C | (A) The set of every output the function actually produces. |
| 2. Domain G | (B) The simplest form of a family, before any shift or stretch. |
| 3. Range A | (C) A rule that sends each input to <i>exactly one</i> output. |
| 4. Average rate of change F | (D) The function that reverses the input-output pairs of f . |
| 5. Concave up H | (E) Feeding the output of one function into another: $f(g(x))$. |
| 6. Parent function B | (F) The slope of the secant line, $\frac{f(b) - f(a)}{b - a}$. |
| 7. Composition E | (G) The set of every input the function is allowed to take. |
| 8. Inverse function D | (H) Bending upward — the rate of change is increasing. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- A relation fails to be a function exactly when (A) two inputs share an output (B) **one input has two outputs** ← (C) the graph crosses the x -axis (D) the domain is restricted
- If a function is decreasing and concave up on an interval, its output is (A) falling faster and faster (B) **falling, but more and more slowly** ← (C) rising more and more slowly (D) changing at a constant rate
- A table of f has equally spaced inputs and **constant second** differences. The best model is (A) linear (B) **quadratic** ← (C) exponential (D) none of these
- The graph of $g(x) = f(x + 3)$ is the graph of f shifted (A) 3 right (B) **3 left** ← (C) 3 up (D) 3 down
- If $f(x) = x^2$ and $g(x) = x - 1$, then $(f \circ g)(4)$ equals (A) **9** ← (B) 15 (C) 16 (D) 3
- A function has an inverse that is also a function exactly when the function is (A) increasing somewhere (B) **one-to-one** ← (C) concave up (D) defined at 0

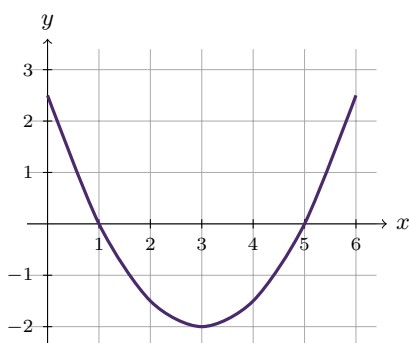
Teacher Note

Item 1: students who pick (A) are confusing the vertical line test with the horizontal one — two inputs *may* share an output. Item 4: the inside change $x + 3$ moves the graph the “wrong” way; expect (A) as the common miss. Item 5: watch for students who compute $g(f(4)) = 15$ instead of $f(g(4)) = f(3) = 9$.

Part C — Short Answer & Computation (30 pts)

Show your work.

- Let $f(x) = 2x^2 - 5$. Compute (a) $f(3)$, (b) $f(-2)$, (c) the value of x (there are two) for which $f(x) = 3$.
(a) $f(3) = 2(9) - 5 = 13$. (b) $f(-2) = 2(4) - 5 = 3$. (c) $2x^2 - 5 = 3 \Rightarrow 2x^2 = 8 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$.
- The graph of $y = f(x)$ is shown below. (a) On what interval is f decreasing? (b) Is f concave up or concave down on the whole picture? (c) List the zeros of f . (d) What is $f(3)$?



(a) Decreasing on $0 < x < 3$ (increasing on $3 < x < 6$). (b) Concave up everywhere shown — the curve bends upward and the rate of change is increasing. (c) Zeros at $x = 1$ and $x = 5$. (d) $f(3) = -2$ (the low point).

- A tank is draining. The volume V (gallons) after t minutes is given below.

t (min)	0	4	8	12
V (gal)	180	148	124	108

- Find the average rate of change of V from $t = 0$ to $t = 8$. (b) Give its units and say in one sentence what it means for the tank. (c) Is the tank draining faster or slower as time goes on? Justify with the numbers. **(a) $\frac{V(8) - V(0)}{8 - 0} = \frac{124 - 180}{8} = \frac{-56}{8} = -7$. (b) Gallons per minute: over the first 8 minutes the tank lost water at an average of 7 gallons each minute (negative because the volume is falling). (c) Slower. The change over each 4-minute block is -32 , then -24 , then -16 gallons — the drops are shrinking, so the draining is slowing down.**
- The table below has equally spaced inputs. Compute the first differences and the second differences, then name the function type (linear, quadratic, or neither).

x	0	1	2	3	4
$g(x)$	2	5	12	23	38

First differences: 3, 7, 11, 15 (not constant, so not linear). Second differences: 4, 4, 4 — constant. The function is quadratic.

5. The parent function is $f(x) = \sqrt{x}$, with domain $[0, \infty)$ and range $[0, \infty)$. Let $g(x) = -2f(x-1) + 4$. **(a)** Describe every transformation, in order. **(b)** State the domain and range of g . **(a)** $g(x) = -2\sqrt{x-1} + 4$: **shift right 1, then vertical stretch by a factor of 2, then reflect over the x -axis, then shift up 4.** **(b)** Domain: $x-1 \geq 0$, so $[1, \infty)$. Range: $\sqrt{x-1} \geq 0$, so $-2\sqrt{x-1} \leq 0$ and $g(x) \leq 4$ — **the range is $(-\infty, 4]$.**
6. Let $f(x) = 3x + 1$ and $g(x) = x^2 - 2$. **(a)** Compute $f(g(2))$. **(b)** Compute $g(f(2))$. **(c)** Write a formula for $(f \circ g)(x)$. **(d)** What do (a) and (b) show about the order of composition? **(a)** $g(2) = 4 - 2 = 2$, so $f(g(2)) = f(2) = 7$. **(b)** $f(2) = 7$, so $g(f(2)) = 49 - 2 = 47$. **(c)** $(f \circ g)(x) = 3(x^2 - 2) + 1 = 3x^2 - 5$. **(d)** $7 \neq 47$: **order matters. Composition is not commutative — $f(g(x))$ and $g(f(x))$ are generally different functions.**
7. Let $f(x) = \frac{x-5}{2}$. **(a)** Find $f^{-1}(x)$ using the swap-and-solve method. **(b)** Verify your answer by computing $f(f^{-1}(x))$. **(a)** Write $y = \frac{x-5}{2}$, swap: $x = \frac{y-5}{2}$, solve: $2x = y - 5 \Rightarrow y = 2x + 5$. So $f^{-1}(x) = 2x + 5$. **(b)** $f(f^{-1}(x)) = \frac{(2x+5)-5}{2} = \frac{2x}{2} = x$. ✓

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- A rectangular dog run is built against a barn wall, so only **three** sides need fencing. The builder has 60 feet of fence. Let x be the length of each of the two sides perpendicular to the barn. **(a)** Write the area A as a function of x . **(b)** What type of function is A , and how do you know? **(c)** State the domain restriction that the context forces, and explain why. **(d)** State one assumption this model rests on. **(a)** **The two perpendicular sides use $2x$ feet, so the side parallel to the barn is $60 - 2x$. Then $A(x) = x(60 - 2x) = 60x - 2x^2$.** **(b)** **Quadratic — the highest power of x is 2, and equally spaced x -values would give constant second differences. It opens downward, so it has a maximum.** **(c)** $0 < x < 30$: a side length must be positive, and the barn side $60 - 2x$ must also be positive, which forces $x < 30$. Outside that window the “rectangle” has no meaning. **(d)** Any one of: no fence is used along the barn; the barn wall is at least $60 - 2x$ feet long; the run is exactly rectangular; no fence is lost to a gate or to overlap.
- Let $f(x) = x^2$ with domain all real numbers. **(a)** Explain why f has no inverse function on that domain. **(b)** Give a domain restriction that fixes the problem, and write the resulting inverse. **(c)** Explain how the graph of a function and the graph of its inverse are related. **(a)** **f is not one-to-one: $f(2) = f(-2) = 4$. Reversing the pairs would send the input 4 to two different outputs, 2 and -2 , and that is not a function. (Equivalently, f fails the horizontal line test.)** **(b)** **Restrict the domain to $x \geq 0$. On $[0, \infty)$ the function is one-to-one, and $f^{-1}(x) = \sqrt{x}$ with domain $[0, \infty)$.** **(c)** **They are reflections of each other over the line $y = x$: the point (a, b) on f becomes (b, a) on f^{-1} , because the inverse swaps input and output.**

Teacher Note

Scoring Part D (5 pts each). Award full credit only when the student both gives the correct result and justifies it. D1: 2 pts (a) $A(x) = 60x - 2x^2$ with the $60 - 2x$ side reasoned out; 1 pt (b) quadratic *with* a reason; 1 pt (c) $0 < x < 30$ tied to the context, not just stated; 1 pt (d) any defensible assumption. A student who writes $2x + 2y = 60$ (all four sides) has missed the barn — give partial credit for correct method on a wrong perimeter. D2: 2 pts (a) not one-to-one *with* a counterexample pair; 2 pts (b) restriction *and* $f^{-1}(x) = \sqrt{x}$; 1 pt (c) reflection over $y = x$. Full-credit language for the arc: *one-to-one* $\rightarrow$ *reverse the pairs* $\rightarrow$ *restrict to make it work* $\rightarrow$ *reflect over $y = x$.*